

Full pumping-cycle optimisation

One periodic path, one nonlinear program, and how to seed it

Transcription, seed generation and staging, with LEI-V3 results

Oriol Cayon

Delft University of Technology

October 7, 2026

Abstract

AWETrim optimises a whole pumping cycle of a soft kite (reel-out figure-eights, the reel-in and both transitions) as *one* periodic phase. The flight path is a single periodic cubic B-spline on the sphere, the depower input is a free per-node profile, the reel speed is a rate-limited control, and the quasi-steady force balance of the reduced-order model (ROM) is imposed at every node. The nonlinear program (NLP) that results is large, stiff and has a nearly flat objective. Whether it converges depends almost entirely on the initial guess. This note documents the transcription, and above all the *seed*: a synthetic cycle that is trim-feasible at every node, closed in tether length, inside every bound and under a curvature limit before the optimiser sees it. Two reel-in constructions are described: a “giant lobe” of the figure-eight and a designed Dubins path on the sphere. Both are followed by a spline design and fairing step, a forward quasi-steady simulation and a rule-based tuner that maps each symptom to the knob that relieves it. A proximal trust-region staging then carries the seed to the optimum in one solve. Results for the LEI-V3 illustrate the pipeline.

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1 Purpose and scope

A pumping cycle has two phases with conflicting requirements. The reel-out flies crosswind figure-eights at high tension; the reel-in depowers the kite, parks it high in the wind window and pulls it back at low tension. The usual treatment optimises them as separate phases joined by boundary conditions. That needs a transition model and hand-chosen switching states, and it splits the cycle-average power into two objectives that are coupled only through those states. AWETrim instead writes the cycle as *one* periodic phase:

- the path is one periodic B-spline over the cycle, so the transitions are just parts of the path, with no switching states;
- depower $u_p(s)$ is a per-node decision over the whole cycle, so *where* to depower is optimised, not prescribed;
- the reel speed v_r is a control and the tether length closes exactly over the period.

The price is an NLP whose feasible set is narrow. At every node, the quasi-steady trim must have a solution with steering and angle of attack inside their bounds, and the shape of the path decides whether it does. The objective, cycle-average power, is a flat ridge: many shapes give almost the same power. Started from a careless guess, IPOPT either fails to find a feasible point or slides along the ridge into a region where the trim no longer exists. Most of the engineering is therefore in the *seed*, and Sect. 3 is the core of this note.

The physics of the ROM is that of Cayon, van Deursen and Schmehl [1]; the trajectory-optimisation framing follows Cayon and Schmehl [2]. The code lives in `scripts/reduced-order-model/optimization/cycle/` (`fit_periodic_cycle.config.py`, `run_full_cycle_opti.py`) and `src/awetrim/kinematics/parametrized_patterns.py` (path constructions, spline design and fairing) and `src/awetrim/timeseries/phase_parametrized.py` (the NLP transcription).

2 Problem formulation

2.1 Quasi-steady kite on a prescribed path

The kite is a point mass at radius r , azimuth ϕ and elevation β in the wind reference frame, flying in the course-aligned reference frame of [1]. The path is prescribed on the unit sphere by a path parameter $s \in [0, 1)$,

$$\mathbf{q}(s) = \begin{pmatrix} \cos \phi(s) \cos \beta(s) \\ \sin \phi(s) \cos \beta(s) \\ \sin \beta(s) \end{pmatrix}, \quad \mathbf{r} = r \mathbf{q}(s), \quad (1)$$

and the kite moves along it at rate $\dot{s}$. With the arc rate $\sigma'(s) = |\mathbf{q}'(s)| = \sqrt{\beta'^2 + \cos^2 \beta \phi'^2}$, the tangential speed and the course angle follow from the path alone,

$$v_\tau = r \sigma'(s) \dot{s}, \quad \chi = \text{atan2}(\phi' \cos \beta, \beta'), \quad (2)$$

and the course rate is $\dot{\chi} = k_g v_\tau / r$, with k_g the geodesic curvature of the path on the unit sphere. In the quasi-steady ROM the tangential acceleration vanishes ($\dot{v}_\tau = 0$) and the force balance in the course frame,

$$\mathbf{F}(\dot{s}, u_s, T, v_r, r, u_p; s) = m\mathbf{a} - \mathbf{F}_{\text{aero}} - \mathbf{F}_{\text{tether}} - \mathbf{F}_{\text{grav}} = \mathbf{0} \in \mathbb{R}^3, \quad (3)$$

is a set of three algebraic equations at every point of the path. Given the state (s, r, v_r) and depower u_p , Eq. (3) determines the speed along the path $\dot{s}$, the steering u_s that flies the prescribed course rate, and the tether tension at the ground T . This is the *trim*. A path is flyable only if this root exists with u_s and the angle of attack α within bounds.

Curvature. The curvature of the path on the unit sphere is $\kappa = |\mathbf{q}' \times \mathbf{q}''|/|\mathbf{q}'|^3$; it splits into a normal part equal to one and the geodesic part, $\kappa^2 = 1 + k_g^2$. The physical curvature of the flown path at radius r is κ/r , and the turn radius is $R = r/k_g$. Because the curvature limits used below correspond to radii of 12–60 m at $r \approx 240$ m ($k_g \gtrsim 4$), κ and k_g differ by less than 3 % there.

2.2 One periodic path for the whole cycle

Azimuth and elevation are uniform periodic cubic B-splines with M control points each,

$$\phi(s) = \sum_{j=0}^{M-1} C_{\phi,j} B_3(Ms - j), \quad \beta(s) = \sum_{j=0}^{M-1} C_{\beta,j} B_3(Ms - j), \quad (4)$$

with the index of B_3 taken modulo M , so the path is C^2 and exactly periodic in s . One period is one whole pumping cycle. The control points $\mathbf{C}_\phi, \mathbf{C}_\beta$ are design variables, and their box bounds bound the path through the convex-hull property. Each basis function is non-zero on four knot intervals only, which the NLP exploits (Sect. 4.2).

2.3 Winch

Two winch descriptions are used, one in the forward simulation that builds the seed and one in the NLP.

Force law (seed simulation). The forward simulation needs a closure for v_r . It uses a linear force law whose zero-force reel speed shifts with depower,

$$T = k(v_r - v_0(u_p)), \quad v_0(u_p) = v_{00} + g(u_p - u_{\text{ref}}), \quad (5)$$

softly clamped to $[T_{\min}, T_{\max}]$ by a softplus/softminus pair of sharpness 10^{-3} N^{-1} . With a negative gain g the *same* law reels out when the kite is powered and reels in when it is depowered (Fig. 1); this is what lets a single phase march through a whole cycle. On the LEI-V3 the law is regressed from the 2019-10-08 flight: $k = 5381 \text{ N s/m}$, $v_{00} = 0.412 \text{ m/s}$, $g = -10.93 \text{ m s}^{-1}$ per metre of power tape around $u_{\text{ref}} = 1.7 \text{ m}$, clamps $[865, 8400] \text{ N}$ (the upper clamp is the drum rating from `system.yaml`).

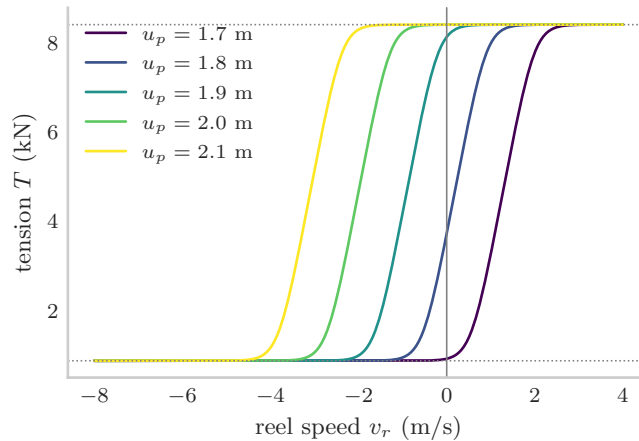


Figure 1: Depower-shifted winch law, Eq. (5), across the identified depower band $u_p \in [1.7, 2.1] \text{ m}$ (power-tape length). The same law holds the reel-out tension when powered and reels in when depowered. Dotted: soft clamps.

Free reel speed (NLP). In the NLP the force law is dropped (`winch_mode: free_speed`). The reel speed is a direct control, rate-limited by the drum acceleration ($|\dot{v}_r| \leq 4 \text{ m/s}^2$) and bounded to $v_r \in [-8, 3.5] \text{ m/s}$, and the tension is only required to lie in the band the drum can hold, $T_{\min} \leq T \leq T_{\max}$. Keeping the force law as an equality has two costs. Its soft clamp has $\partial T / \partial v_r \approx 0$ on the plateau, which removes rank from the Jacobian exactly where the optimum rides the force limit. It also ties radial closure to the tension through a stiff relation. With free speed, closure is nearly linear in a direct control. The optimal $T(v_r, u_p)$ relation is an *output*, from which a winch controller can be regressed afterwards (Fig. 8).

2.4 Transcription

The cycle is discretised on N nodes $s_i = i/N$. Per node the decisions are

$$\mathbf{x}_i = (\dot{s}_i, u_{s,i}, u_{p,i}, v_{r,i}, r_i, T_i), \quad i = 0, \dots, N-1, \quad (6)$$

and the design parameters are $\mathbf{p} = (\mathbf{C}_\phi, \mathbf{C}_\beta, r_0)$, where the operating radius r_0 is free within 180–300 m. Time follows from the path, $\Delta t_i = \Delta s / \dot{s}_i$ (left rule, the same quadrature as the simulation), and the objective is the cycle-average mechanical power

$$\max_{\mathbf{x}, \mathbf{p}} \bar{P} = \frac{\sum_{i=0}^{N-2} T_i v_{r,i} \Delta t_i}{\sum_{i=0}^{N-2} \Delta t_i}. \quad (7)$$

Table 1 lists the constraints. Two of them define the cycle: the radial kinematics $r_{i+1} = r_i + v_{r,i} \Delta t_i$ and the hard closure $r_{N-1} = r_0$. Closure must be an *equality*. As a penalty it lives in the objective, where it is exact only as its weight goes to infinity, and the power reward for reeling out forever beats any finite weight: the optimiser opens the cycle, the radius runs away and the trim breaks.

Table 1: Constraints of the full-cycle NLP (per node unless stated). Bounds are `DEFAULT_OPTI_LIMITS`, overlaid with the hardware limits of `system.yaml` and the seed’s `opti_limits_override`; values are those of the LEI-V3.

constraint	expression	role
trim	$\mathbf{F}(\mathbf{x}_i; s_i, \mathbf{p}) = \mathbf{0}$ (3 rows)	Eq. (3), ROM
radial kinematics	$r_{i+1} = r_i + v_{r,i} \Delta s / \dot{s}_i$	tether length
radius pin, closure	r_0 free, $r_{N-1} = r_0$	periodic cycle
winch band	$T_{\min} \leq T_i \leq T_{\max}$	drum force rating
reel speed	$-8 \leq v_{r,i} \leq 3.5 \text{ m/s}$	drum speed range
rate limits	$ \Delta u_s / \Delta t \leq 0.145 \text{ s}^{-1}$, $ \Delta u_p / \Delta t \leq 0.2 \text{ m/s}$ $ \Delta v_r / \Delta t \leq 4 \text{ m/s}^2$	KCU drum
steering	$ u_{s,i} \leq 0.175$	KCU range
angle of attack	$0 \leq \alpha_i \leq 14^\circ$	ROM validity
height	$r_i \sin \beta(s_i) \geq 50 \text{ m}$	safety
path speed	$\dot{s}_i \geq 0.01$	$\Delta t_i > 0$
coefficients	$C_\beta \leq 80^\circ$, box on C_ϕ	convex hull
turn radius (opt.)	$r_i \sigma'^3 \geq R_{\min} \kappa \sigma'^3$, $\sigma' \geq 0.2 \bar{\sigma}'$	4 samples/interval

The optional turn-radius constraint is written in polynomial form, $-r\sigma'^3 \leq R_{\min} \kappa \sigma'^3 \leq r\sigma'^3$, rather than as $R_{\min} |\kappa| \leq r$. Near a forming cusp $\kappa \propto \sigma'^{-3}$ blows up, and a cold solve that passes through such a shape stalls on the exploding rows; the cross-product numerator stays bounded. It is enforced on four sub-samples per interval because the spline can fold a cusp *between* two nodes, which no node-wise quantity sees. A floor on the arc rate keeps the spline from “hesitating” ($\sigma' \rightarrow 0$ over a sliver of s) to dodge it.

2.5 What the seed must provide

The NLP is solved with IPOPT from a primal warm start, the forward simulation of the seed. Experience with this problem gives five requirements:

1. **Trim everywhere.** Every node of the seed must trim. A node that fails leaves the per-node root problem without a nearby solution, and padding the warm start there breaks the NLP in ways that are hard to trace (`require_full_trajectory` aborts instead).
2. **Inside the bounds.** Steering and angle of attack must stay within their bounds, so the active set at the start is the right one; a small steering overshoot (0.05) is tolerated, because the flown KCU itself saturates at the bound and the NLP clips those nodes.
3. **Closed.** $|r_{\text{end}} - r_0| \leq 10 \text{ m}$; an open seed starts the hard closure row far from feasible.
4. **Smooth.** Path curvature under $\kappa_{\text{max}} = 0.08 \text{ m}^{-1}$ (radius 12.5 m). Sharp handovers between figures and reel-in show up as steering spikes the actuator cannot follow.
5. **The right topology.** The optimiser can reshape the path, but it does not change the number of loops or the side of the reel-in reliably. Those choices belong to the seed.

Section 3 constructs a seed that meets all five without any flight data.

3 Seed generation

3.1 Overview

Figure 2 summarises `fit_periodic_cycle.config.py`. Everything kite-specific comes from the kite’s folder: hardware and actuator ranges from `system.yaml`, the ROM and its identified depower band from the ROM configuration, and wind, winch law and pattern size from `cycle_profile.yaml`. Nothing about a particular kite lives in the code. The pipeline is geometric up to the forward simulation; only the tuning loop costs simulations (about one minute each).

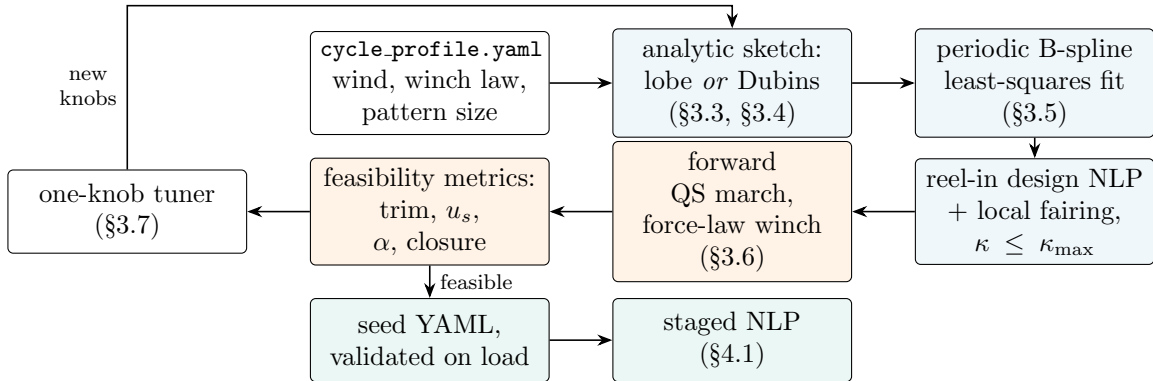


Figure 2: Seed pipeline. Blue: pure geometry; orange: forward simulation; green: written seed and the optimiser. An infeasible seed passes its dominant symptom to the tuner, which moves one knob and rebuilds the sketch; this loop runs only with `--auto`.

3.2 The reel-in window and the synthetic depower

A fraction f of the period is reel-out and the rest is reel-in. The reel-in is a window of half-width $h = (1 - f)/2$ centred at s_c , described by a smooth bump $w(s) \in [0, 1]$ built from Hermite smoothsteps $S(x) = 3x^2 - 2x^3$ on ramps of width $\rho(1 - f)$:

$$w(s) = S\left(\frac{d + h}{\rho(1 - f)}\right) \left[1 - S\left(\frac{d - h + \rho(1 - f)}{\rho(1 - f)}\right)\right], \quad d = ((s - s_c + \tfrac{1}{2}) \bmod 1) - \tfrac{1}{2}, \quad (8)$$

with S clamped to $[0, 1]$. Measuring d circularly lets the window wrap across the periodic seam, so any centre is valid. The *same* bump drives the path shape and the synthetic depower,

$$u_p(s) = u_{p,\text{pow}} + \delta w(s) (u_{p,\text{dep}} - u_{p,\text{pow}}), \quad (9)$$

where $\delta \in (0, 1]$ is the depower *depth*. A mismatch between where the kite is depowered and where it is flown high is a reliable way to break the trim: a kite still powered while parked at 70° elevation has no quasi-steady solution. The depth is the single radial-closure knob. Through Eq. (5) it sets the reel-in speed and therefore how much tether comes back in.

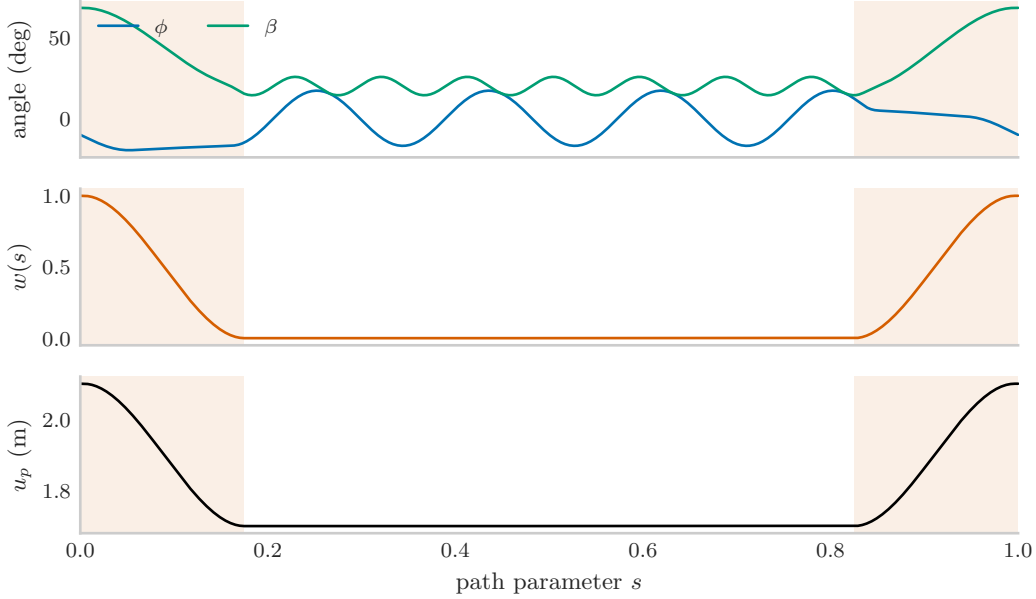


Figure 3: Lobe seed of the LEI-V3 over one period, at the default knobs ($f = 0.65$, ramp fraction $\rho = 0.49$, window centred at $s = 0$, the top of the reel-in). Top: path angles; middle: the reel-in bump w ; bottom: the synthetic depower profile, Eq. (9), with $\delta = 1$. Shaded: reel-in window.

3.3 Lobe reel-in: the cycle as one giant figure lobe

The first construction (`bow_shape="lobe"`, the default) fades the figure-eight out over the window and lifts the elevation:

$$\phi(s) = (1 - w) A \sin \psi(s) + w a b(\xi), \quad (10)$$

$$\beta(s) = \beta_0 + (\beta_{\text{pk}} - \beta_0) w + (1 - w) B \sin 2\psi(s), \quad (11)$$

with figure amplitudes A, B , centre elevation β_0 , reel-in peak β_{pk} , azimuth bow a and $\xi \in [0, 1]$ the position within the window. Three details make this a feasible reel-in rather than a sketch:

Frozen phase. Over the first part of the entry ramp the figure rate $\dot{\psi}$ fades to zero with a smoothstep, so ψ advances only a small budget (0.5 rad) and then *holds* through the window, resuming symmetrically on the exit ramp. The fading figure term is then a constant offset rather than a wiggle, and the climb peels off the last centre crossing in one clean arc. The figure-count residual that makes the phase exactly periodic is hidden in the plateau, where the figure amplitude is exactly zero.

Pinned handovers. The frozen phases are pinned: $\psi_{\text{entry}} = \pi - 0.3$ (just before the climbing centre crossing) and $\psi_{\text{exit}} = 3\pi/2$ (the lobe extreme, heading down). The descent then lands *tangentially* on a lobe and flies its lower half into the next crossing, so the reel-in replaces the top half of one lobe and slots into the alternation. The kite does not come down on a lobe that is already in progress.

One-sided bow. The azimuth bow $b(\xi)$ rises over the last 30 % of the entry ramp and does not return. Wherever the elevation plateau is flat, the azimuth must carry the path speed; a bow that stayed zero across the top would stall the parametrisation into a cusp.

Counting loops. With both phases pinned, the reel-out flies $n_{ro} = k + \gamma$ figures, where k is an integer and γ the fractional figure fixed by the phase gap. The number of half-figures *visible* during reel-out is therefore $2n_{ro}$, and its parity decides the re-entry side: odd counts re-enter opposite the peel-off lobe, even counts on the same side. The user-facing knob `--loops` is the visible count. The generator inverts the snap and, when the requested parity does not match, moves the peel-off to the mirrored climbing crossing ($\psi_{\text{entry}} + \pi$) without touching the landing. The one free topological choice is `--reelin smooth|cross`: whether the top loop goes straight to the landing side or crosses the ascending leg lower down.

3.4 Dubins reel-in: a designed path on the sphere

The lobe shape fades one curve into another and inherits whatever curvature the fade produces. The second construction (`--reelin dubins`) designs the reel-in instead. The reel-out flies the plain figure $\phi = A \sin \psi$, $\beta = \beta_0 + B \sin 2\psi$ from a landing at $\psi = 3\pi/2$ over a prescribed number of visible half-figures to a peel-off state, and the reel-in joins the peel-off state (position *and* heading) to the landing state through a level apex at β_{pk} .

Motion at bounded curvature on the sphere. Let $F = [\mathbf{q}, \mathbf{t}, \mathbf{b}]$ be the orthonormal frame of position, heading and $\mathbf{b} = \mathbf{q} \times \mathbf{t}$. Per unit arc length on the unit sphere, $\mathbf{q}' = \mathbf{t}$, $\mathbf{t}' = -\mathbf{q} + k\mathbf{b}$, $\mathbf{b}' = -k\mathbf{t}$, i.e. $F' = F\hat{\omega}$ with the constant axis $\hat{\omega} = (k, 0, 1)$. A piece of constant geodesic curvature k is therefore a rotation in closed form (Rodrigues),

$$F(\sigma) = F_0 \exp(\sigma\hat{\omega}) = F_0 \left[\mathbb{I} + \sin \theta \hat{\mathbf{a}} + (1 - \cos \theta) \hat{\mathbf{a}}^2 \right], \quad \mathbf{a} = \frac{\hat{\omega}}{|\hat{\omega}|}, \quad \theta = \sigma \sqrt{1 + k^2}, \quad (12)$$

with $k = 0$ a great circle. A turn of radius R at r_0 has $k = r_0/R$.

Shortest legs. Each leg (peel-off→apex, apex→landing) is the shortest arc–straight–arc (CSC) path between the two frames: arcs at $\pm k$, a great circle between them, i.e. a Dubins path [3] on the sphere [4]. The three lengths of each of the four types (LSL, RSR, LSR, RSL) are found by multi-start least-squares shooting on the exact motion; the shortest converged solution is kept per type.

Apex and radius are searched, not set. The apex azimuth is scanned over 25 values in $\pm 1.2|\phi_{\text{land}}|$ and the shortest whole reel-in is kept. A prescribed apex near the climb admits only paths with an extra loop, and at high elevation a sideways metre is a lot of azimuth. Candidates that rise above the apex or wrap behind the ground station are rejected. Radii from the default 60 m down to the kite’s turn-radius floor (or half the default) are tried in steps of $\times 0.85$. The reel-in with the least *total turning* wins, the larger radius on a tie, so a tighter radius is used only where it saves a detour loop.

Timing. The reel-in occupies the window $u \in [f, 1]$ of the period. Its path speed eases from the figure speed at each end (over at most 15 % of the window) to a constant middle speed that covers the path length exactly, so the curve is C^1 in s and the periodic spline fits it without ringing.

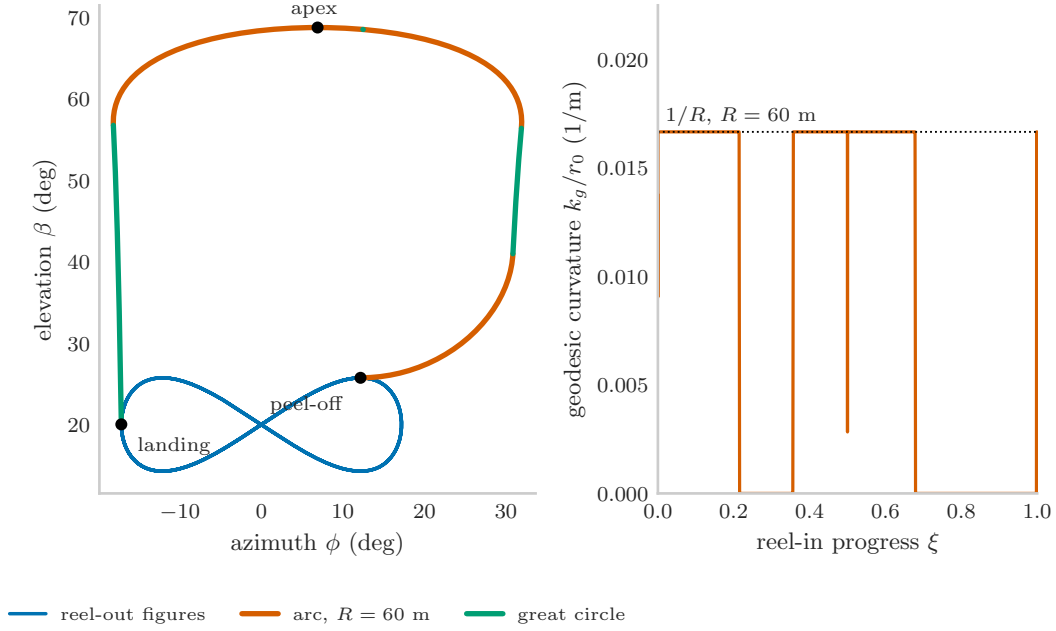


Figure 4: Dubins reel-in of the LEI-V3 seed, 7 visible half-figures, apex at $\beta_{\text{pk}} = 68.8^\circ$. Left: path in the wind window, reel-in coloured by piece. Right: geodesic curvature along the reel-in: arcs at $R = 60$ m, great circles at zero. The search keeps 6 pieces (two CSC legs), a reel-in of 462 m at $r_0 = 236.7$ m.

3.5 From sketch to spline

Fit. The analytic sketch is sampled at 600 points and fitted by linear least squares with the periodic basis of Eq. (4). The knots are uniform in s , so the local resolution can only be set through M . The generator takes

$$M = \max\left(40, \lceil 10 n_{\text{loops}} \rceil, \lceil 15/(1-f) \rceil\right), \quad (13)$$

that is, at least 10 control points per figure-eight and 15 across the reel-in window. These are the measured knees of a convergence study (≤ 1 m path deviation at r_0). An under-resolved fit *rings*: the fitted spline’s curvature can be several times that of the curve it fits.

Designing the reel-in. If the fitted spline violates κ_{max} , the reel-in sketch is treated as a drawing only, and the reel-in is designed between the two frozen handover states. The control points whose whole support lies inside the window, $\Delta \mathbf{c}$, solve

$$\begin{aligned} \min_{\Delta \mathbf{c}} \quad & \frac{\sum |\mathbf{q}''|^2}{E_{b,0}} + w_t \frac{\sum |\mathbf{q}'|^2}{E_{t,0}} + w_{\text{pk}} (\beta(s_{\text{apex}}) - \beta_{\text{pk}})^2 + w_{\text{tr}} \frac{\sum |\Delta \phi|^2 + |\Delta \beta|^2}{n (0.1)^2} \\ \text{s.t.} \quad & \kappa^2 \leq (0.97 \kappa_{\text{max}} r_0)^2, \quad |\mathbf{q}'|^2 \geq \frac{1}{4} \min |\mathbf{q}'_0|^2, \quad |\Delta \mathbf{c}| \leq 0.5, \end{aligned} \quad (14)$$

with sums over a dense grid (4000 samples). The objective is a spline in tension: bending energy plus a Dirichlet term ($w_t = 1$). Without the tension term, the minimum-bending way round a too-sharp corner is an outward teardrop loop. The peak elevation is a target to hit ($w_{\text{pk}} = 100$), never a knob that is quietly lowered. A weak tracking of the sketch ($w_{\text{tr}} = 0.1$) holds the topology (which side, crossing or tangential). The window-edge control points stay frozen, so the handover states and everything outside the window are bit-identical, and C^2 continuity is automatic. The curvature rows are normalised to $O(1)$ ($\kappa^2/(\cdot)^2 - 1 \leq 0$) because the raw form spans orders of magnitude across samples. The floor on $|\mathbf{q}'|$ prevents a subtle failure: without it the solver satisfies the bound by stalling the parametrisation between samples, and a forming cusp hides between grid points.

Fairing. A residual pass catches anything outside the designed window (figures, window edges). It finds the samples above $0.97\kappa_{\max}$ and frees the control points supporting them plus one neighbour per side. It then moves them by the *least* amount, $\min|\Delta\mathbf{c}|^2$, under the same curvature and arc-rate rows, inside a cumulative box of 0.2 rad. When a round fails it widens the freed set (up to five rounds). The box is essential: from a badly violating start, IPOPT otherwise wanders into a curvature-feasible but wildly deformed basin of giant loops.

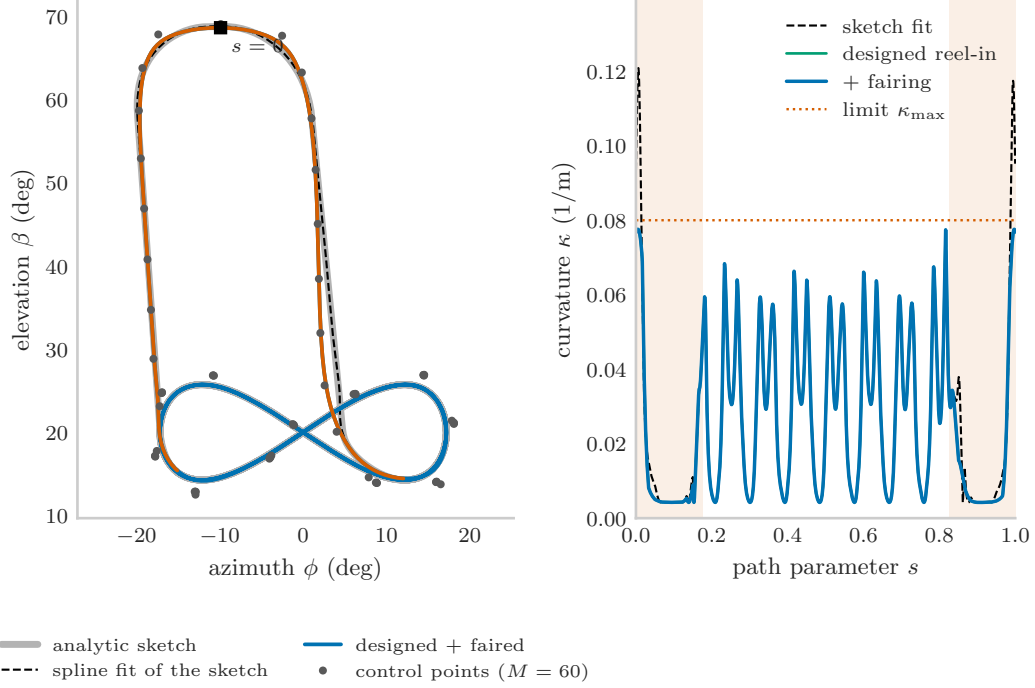


Figure 5: Lobe seed of the LEI-V3 at the default knobs ($M = 60$). Left: analytic sketch, its spline fit, and the designed + faired spline (reel-in in orange). Right: curvature along the path. The fitted sketch peaks at 0.121 m^{-1} at the reel-in handovers; the reel-in design (17 control points) brings it to 0.078 m^{-1} while hitting the peak elevation to within $68.6^\circ/68.8^\circ$ and moving the path by at most 2.1° . The fairing then has nothing left to do (0 points moved).

3.6 Forward simulation

The geometric seed is then flown. The forward simulation marches node by node along the spline. At each node it solves the trim, Eq. (3), together with the winch law, Eq. (5), for $(\dot{s}, u_s, T, v_r)$ (a 4×4 root problem), and it integrates r with the same left rule as the NLP. Each node is seeded from the previous one, with tiered fallback guesses when the first attempt fails. The march starts in steady reel-out at the maximum tension and r_0 . Its result is the NLP’s warm start, and it is reused for the first stage, so the optimiser does not repeat the march. The measured cycle duration then sets the NLP grid, $N = \lceil 3T_{\text{cycle}} \rceil$ nodes. The duration prior in the profile can be off by a factor of two at other winds or loop counts. In a convergence sweep, 3 nodes/s kept the cycle power within 0.2% of a 400-node reference. At 2 nodes/s steering peaks fell between nodes (the NLP only bounds nodes, so a coarse grid hides violations rather than removing them), and at 1 node/s the trim broke.

3.7 Automatic tuning

With `--auto` the generator iterates `simulate` $\rightarrow$ `metrics` $\rightarrow$ *one knob move*. The metrics localise each violation in s , and the location selects the lever:

- a node that fails to trim is fixed first, in its region;

- angle-of-attack violations are always the slow top of the reel-in, where the kite is starved of apparent wind: lower β_{pk} by 0.1 rad;
- steering beyond the slack is handled according to where it occurs (Table 2);
- an open cycle is closed by the depower depth, $\delta \leftarrow \delta + \Delta r / (40 \text{ m})$ (measured $\partial \Delta r / \partial \delta \approx -40 \text{ m}$), and, once fully depowered, by shortening the reel-out fraction f .

Table 2: Lever chains of the tuner by the region of the worst violation. A symptom that survives a move without improving the score by 10 % escalates to the next lever in its chain.

region	levers, in order	reason
reel-in top	widen bow $ a $ (+0.06 rad); lower β_{pk}	wider, faster arc at low wind
reel-in edge	gentler ramp ρ ; bigger figures; lower β_{pk}	handover sharpness
figure-eights	bigger figures ($A \times 1.12$, $B = A/2.5$)	turn rate $\sim 4\pi/T_{loop}$

The figure-eight lever comes from a simple estimate. A figure-eight sweeps the course angle by $\pm 2\pi$ per lobe, so at the kite’s natural speed the steering demand scales with the turn rate $4\pi/T_{loop}$. The way to turn more gently is a longer arc, i.e. bigger figures; azimuth and elevation amplitudes are scaled together at $A \approx 2.5B$. Each iterate is scored,

$$\mathcal{S} = 100 [\text{trim failed}] \left(2 - \frac{i_{fail}}{N} \right) + 50 \max(\alpha - \alpha_{max}, 0) + 10 \max(|u_s|_{max} - u_{s,max} - 0.05, 0) + \frac{\max(|\Delta r|/m - 10, 0)}{10} + 25 \max\left(\frac{\kappa_{max}^{spl}}{\kappa_{max}} - 1, 0\right), \quad (15)$$

and the *best* iterate is written, not the last (at most 12 iterations). The final knobs are printed so that they can be pinned in `cycle_profile.yaml` and reproduced without tuning. Section 5 shows a trace.

3.8 Validation on load

The optimiser re-checks the seed before solving and refuses it if it is stale, hand-edited or assembled from mixed sources. The checks are: the solver keys the generator writes are present; the depower profile has $N + 1$ entries inside the identified band; and no node in the top tenth of the elevation range has a winch offset $v_0(u_p) > 0$, i.e. the kite is never commanded to reel out while parked high.

4 Optimisation strategy

4.1 Proximal staging

The power objective is a flat ridge, and the periodic spline has near-null re-parametrisation directions. Left free, IPOPT takes huge steps ($\|\mathbf{d}\| \sim 10^2$) along them. Those steps merge or destroy figure-eight loops and strand the iterate outside the trim-feasible region, where the primal infeasibility explodes and never recovers. The solve is therefore regularised with a proximal term on the design parameters,

$$\min -\frac{\bar{P}}{P_s} + w \sum_{p \in \mathbf{p}} \left(\frac{p - \hat{p}}{\Delta_p} \right)^2, \quad (16)$$

where $\hat{p}$ is the warm-start value, Δ_p the width of the parameter’s bound and P_s the seed’s average power. The quadratic anchor adds curvature exactly in the null directions and so bounds the step, while it barely resists directions with a real power gradient. A single solve can therefore travel the whole way. Two stages are run (Table 3): a main stage with $w = 0.05$, then a warm-started polish with $w = 0.005$ that re-anchors at the main optimum and removes most of the bias towards

the seed. A hard box $|p - \hat{p}| \leq \delta_p$ remains as a backstop that should never bind; if it does, the driver warns and a re-centred repeat pass (`--stage-repeats`) or a continuation (`--continue-from`) is the remedy. After each stage a topology signature (the number of prominent azimuth turns, about two per figure-eight plus the reel-in) is compared with the seed's, and a change is reported even when IPOPT reports success.

Table 3: Stages of `run_full_cycle_opti.py`. Both optimise (C_ϕ, C_β, r_0) and all per-node decisions under the hard closure; at most 1000 IPOPT iterations each.

stage	anchor w	backstop box δ_p	IPOPT start
main	0.05	0.15 rad (C), 50 m (r_0)	cold barrier, seed primal
polish	0.005	0.15 rad (C), 100 m (r_0)	warm: $\mu_0 = 10^{-4}$, bound push 10^{-6}

4.2 Numerics

Scaling. Tension ($\sim 10^3$ – 10^4 N) and radius ($\sim 10^2$ m) are carried as $O(1)$ decisions divided by the 90th percentile of the warm start, and all equality rows are scaled likewise. Unscaled, the NLP oscillates and falls into restoration.

Local support. A shared node function with symbolic s references all M control points and produces a dense M -wide Jacobian block per node. Instead, node functions are built per knot interval from a truncated copy of the spline that keeps only the four basis terms non-zero there. This is exact on that interval, and each node then couples to 4+4 control points. The Jacobian is about six times sparser, and whole-graph SX expansion costs about 3.7 MB per node instead of 24 MB.

Solver. IPOPT with MUMPS and an L-BFGS Hessian, adaptive barrier, gradient-based scaling, $\text{tol}=10^{-5}$. L-BFGS floors the dual infeasibility at 10^{-2} –1 on this problem, so the acceptable-point test is set to fire on feasibility alone (constraint violation $< 10^{-4}$, objective change $< 10^{-4}$ over five iterations).

5 Results for the LEI-V3

All results are for the LEI-V3 with the flight-calibrated (semi-empirical) ROM, at the wind of its `cycle_profile.yaml`: an explog profile with 6 m/s at 6 m. They come from two different code states, which this section keeps apart:

- **Seeds** (Sect. 5.1) were generated on 2026-10-06 with the ROM rework later merged in PR #22 (standardised steering, bound $|u_s| \leq 0.175$; AoA bound 14°), still on the semi-empirical ROM, i.e. before the aerostructural ROM became the LEI-V3 default.
- **The optimum** (Sect. 5.2) is the last converged LEI-V3 full-cycle solve, from 2026-08-27, at the same wind. It predates the Dubins reel-in, the reel-in design step and the steering standardisation, so it is shown as an illustration of what the staged NLP does, not as a result of the code described above.

Sect. 5.3 states why no fresh optimum is reported.

5.1 Seed generation

Two seeds with 7 visible half-figures were generated. The *lobe* seed used `--auto`, starting from the default knobs of Fig. 5. The *Dubins* seed used `--reelin dubins --loops 7 --check`: no tuning, only the forward simulation and the grid recalibration. Figure 6 shows both paths and Table 4 summarises them.

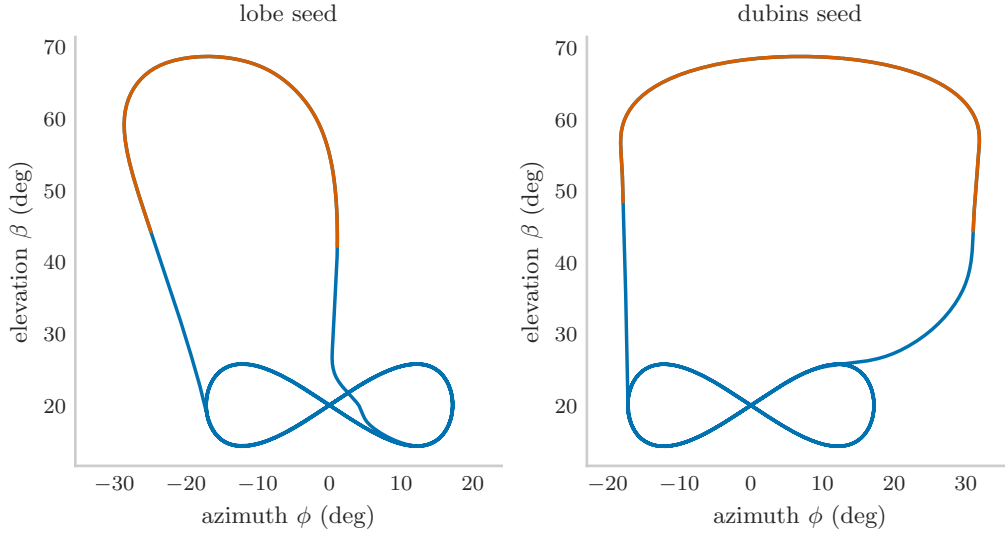


Figure 6: The two generated LEI-V3 seeds in the wind window; reel-in (depower above mid-band) in orange. Left: lobe seed after `--auto`; right: Dubins seed.

Table 4: Generated seeds and their forward simulations. Steering and AoA ranges are to be read against the NLP bounds $|u_s| \leq 0.175$ (plus the 0.05 warm-start slack) and $\alpha \leq 14^\circ$; closure against 10 m.

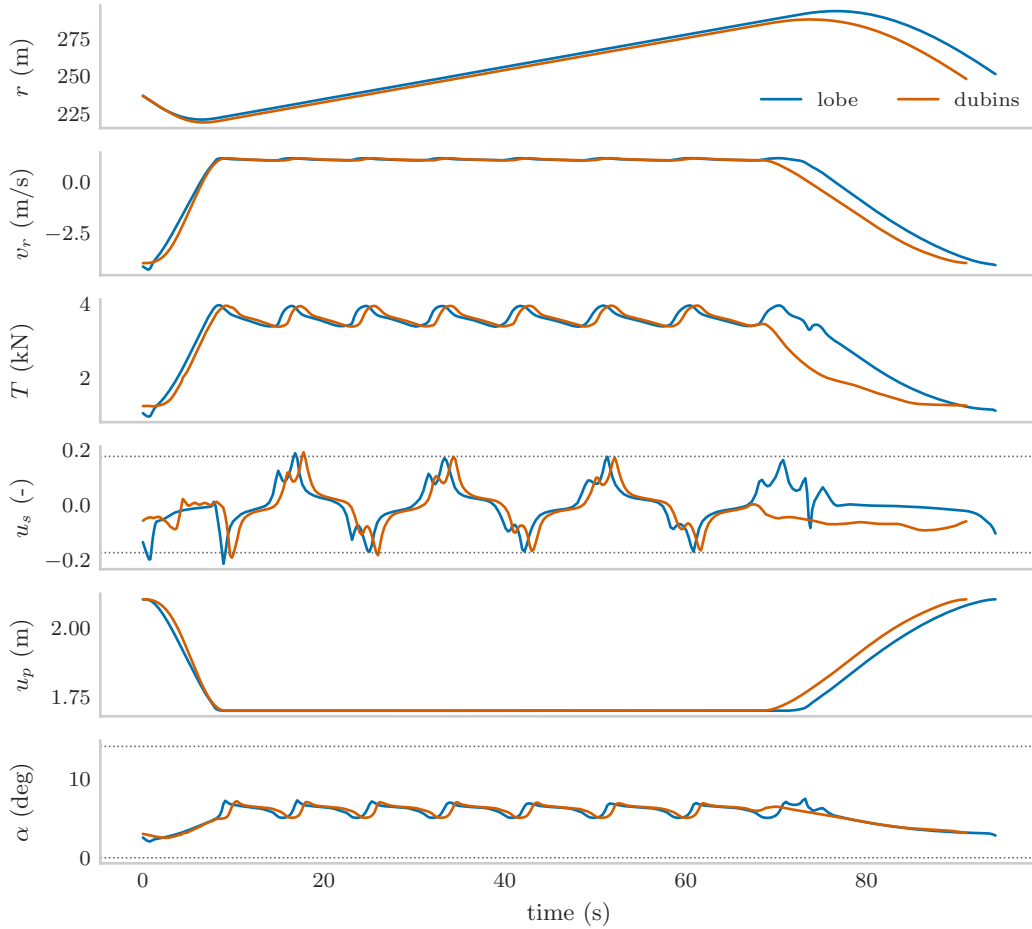
quantity	lobe seed	dubins seed
M / N	60 / 283	70 / 275
nodes trimmed	283/283	275/275
max curvature (1/m)	0.077	0.065
cycle duration (s)	94.2	91.0
average power (kW)	1.83	1.78
closure $r_{\text{end}} - r_0$ (m)	+14.5	+11.2
u_s range (-)	-0.22 / +0.19	-0.19 / +0.19
max α (deg)	7.4	7.1
max tension (kN)	3.96	3.95
tuner iterations	8	—
generation time (min)	16	5

What worked. Both seeds trim at every node of the recalibrated grid, stay under the curvature limit, and keep the angle of attack inside its bound with a wide margin. The Dubins seed reached this *without any tuning*. Its steering grazes the bound $(-0.19/+0.19)$ by less than the slack. The lobe seed needed the tuner (Table 5), and its first three moves are the case the lever chains were written for. The steering violation sat at the top of the reel-in ($s \approx 0.01$ – 0.02 ; the window is centred on $s = 0$), and widening the bow $|a|$ shrank it step by step, from 31 violating nodes to 9.

What did not. Neither seed closes to within 10 m: they end 14.5 m (lobe) and 11.2 m (Dubins) long. For the lobe seed, the tuner’s closure lever was already exhausted at full depower depth, and its fallback, shortening the reel-out fraction from 0.65 to 0.53, moved the gap by less than a metre. The tuner therefore stopped and wrote its best iterate (“cannot close the cycle (gap +15.8 m) within knob ranges”). A closure that is insensitive to the reel-in fraction means that the reel-in speed, and not the reel-in duration, limits how much tether comes back. The reel-in speed is set by the depower-shifted winch law of Eq. (5). On paper, a 11–17 m gap on $r_0 = 236.7$ m is a small closure residual that the NLP’s first iterations can absorb (Sect. 2.5).

Table 5: --auto trace of the lobe seed: one knob per iteration, chosen from the dominant symptom and its location.

it.	dominant symptom	knob move
1	steering $[-0.37, +0.19]$ at 31 node(s), worst at $s=0.02$	az_reelin_amp $-0.36 \rightarrow -0.42$
2	steering $[-0.32, +0.19]$ at 12 node(s), worst at $s=0.01$	az_reelin_amp $-0.42 \rightarrow -0.48$
3	steering $[-0.24, +0.19]$ at 9 node(s), worst at $s=0.01$	az_reelin_amp $-0.48 \rightarrow -0.54$
4	under-reel +14.9 m at full depower	reelout_fraction $0.65 \rightarrow 0.61$
5	under-reel +14.8 m at full depower	reelout_fraction $0.61 \rightarrow 0.57$
6	steering $[-0.26, +0.18]$ at 9 node(s), worst at $s=0.02$	az_reelin_amp $-0.54 \rightarrow -0.60$
7	under-reel +14.5 m at full depower	reelout_fraction $0.57 \rightarrow 0.53$
8	cannot close the cycle (gap +15.8 m) within knob ranges	stop, best iterate

**Figure 7:** Forward simulations of the two seeds, which are the NLP warm starts. Dotted: NLP bounds on steering and angle of attack. Both cycles are about 90 s long; the reel-in is the stretch of negative v_r and high u_p .

5.2 An optimised cycle (2026-08-27)

Figures 8 and 9 show the last converged solve. The figure was written by that run itself and compares its seed (a lobe seed of that time) with the optimum. The staged NLP raised the cycle-average power from 1.24 kW to 4.46 kW and shortened the cycle to 68.6 s, with the closure satisfied to 0.00 m and r_0 moved to 241 m. The mechanisms are visible in the per-node decisions:

- **Reel-out at higher load.** The depower profile leaves the identified band towards the powered side (u_p down to 1.34 m), and the tension rises to 5.95 kN peak (4.42 kN mean),

reeling out at up to 2.2 m/s.

- **Shorter, faster reel-in.** The reel speed reaches -5.0 m/s, and the reel-out time fraction of the cycle is 0.72.
- **Free-speed winch.** The optimal $T(v_r)$ points do not lie on the flight-regressed force law the seed was marched with (Fig. 8, bottom left). The regression of a controller from the optimum is a separate step.

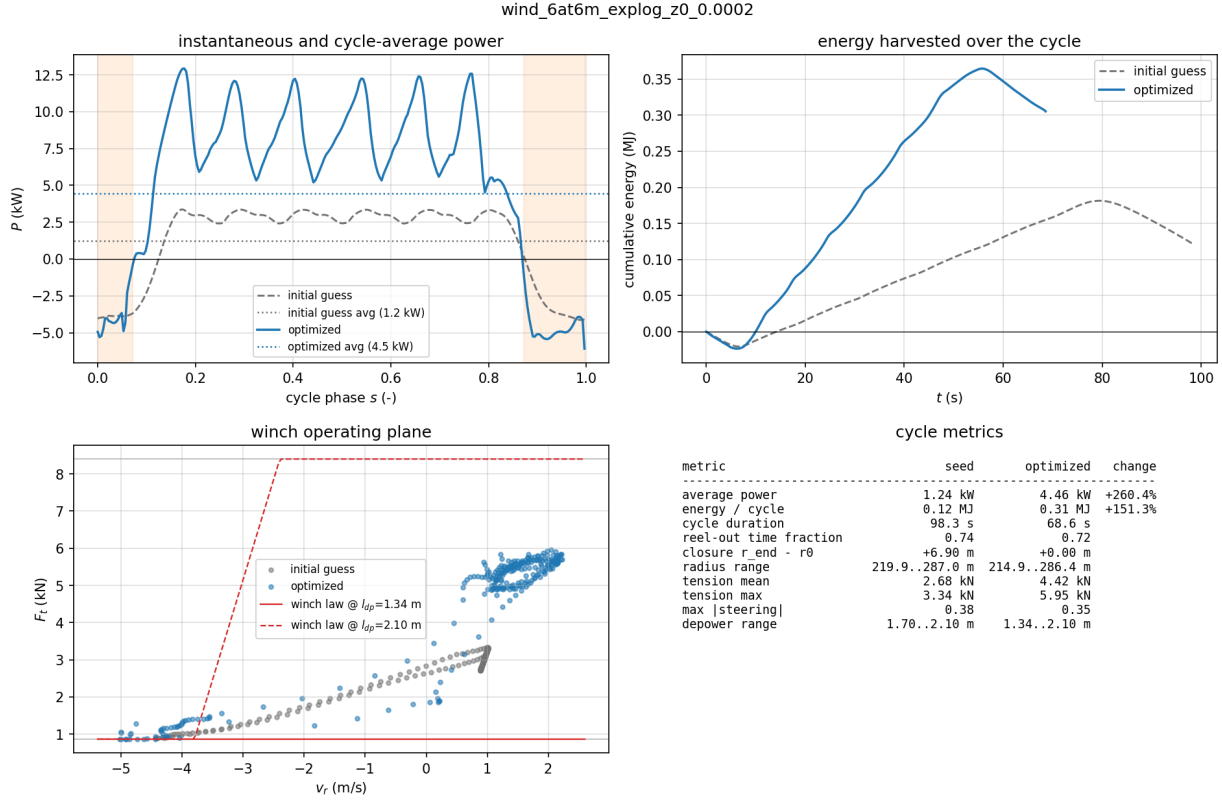


Figure 8: Seed versus optimum of the 2026-08-27 LEI-V3 solve, as written by `run_full_cycle_opti.py`: instantaneous and average power, cumulative energy, winch operating plane and metrics. Shaded: reel-in.

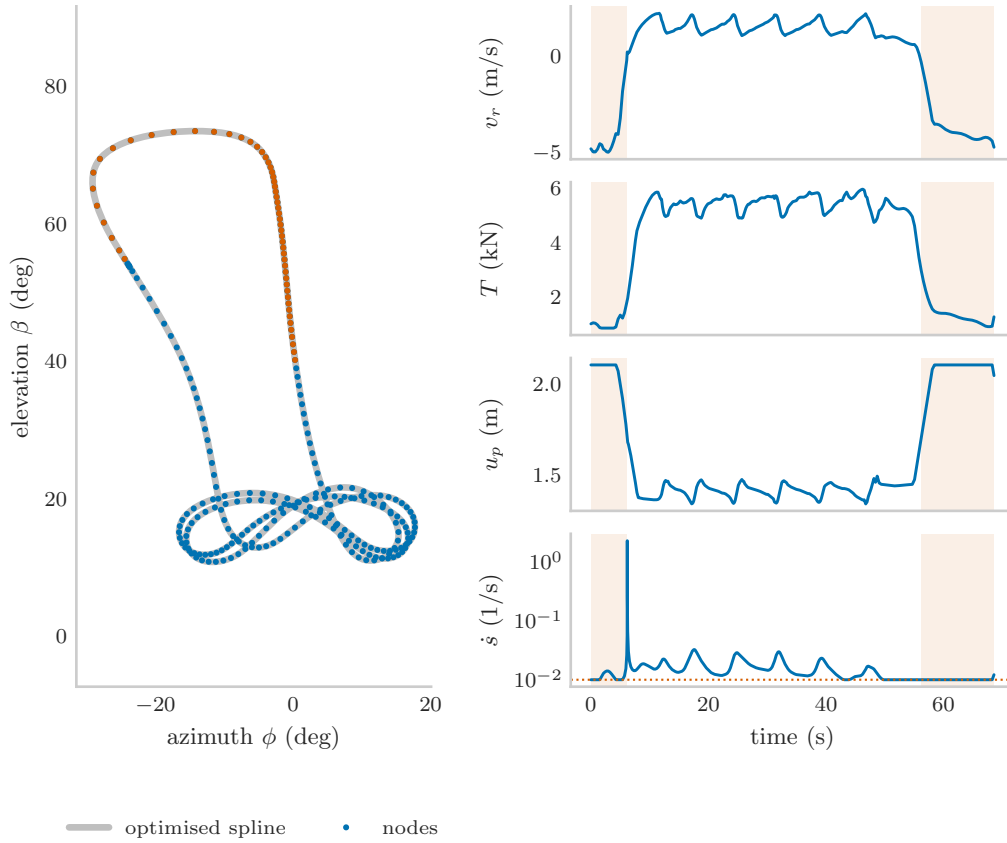


Figure 9: The 2026-08-27 optimum ($M = 43$, $N = 265$ nodes). Left: optimised spline and the node positions, reel-in ($v_r < 0$) in orange. Right: reel speed, tension, depower and path rate $\dot{s}$; the dotted line is the numerical floor $\dot{s} \geq 0.01$.

Caveats of this optimum. Four features of the 2026-08-27 solution show what the constraints of that time did not yet cover. Each motivated a later addition to the method:

1. The depower profile goes down to 1.34 m, below the band $[1.7, 2.1]$ m on which the ROM was identified, so the reel-out extrapolates the ROM. The band is now declared in the ROM file, and the AoA bound can be set from the ROM's validity range.
2. The flown turn radius drops to 10.4 m, because no turn-radius constraint existed. The optional sub-sampled constraint of Table 1 was added for this.
3. 62 of the 265 nodes sit on the numerical floor $\dot{s} = 0.01$: 42 while reeling in, most of the rest in the transition into the reel-in (Fig. 9, bottom right). There the time step $\Delta t = \Delta s / \dot{s}$ is capped by a guard rather than set by the physics, so the reel-in timing of this optimum is partly an artefact of that bound.
4. One node ($s = 0.083$) has $\dot{s} = 2.27 \text{ s}^{-1}$, about 150 times the median 0.0149 s^{-1} . Since $v_r = r\sigma'\dot{s}$ stays ordinary, the arc rate σ' of the spline nearly vanishes there and $\dot{s}$ jumps to compensate, a *hesitation point*. The floor $\sigma' \geq 0.2\bar{\sigma}'$ that now comes with the turn-radius constraint (Table 1) excludes it.

Its steering also reaches $|u_s| = 0.35$, which was the steering bound before the steering input was standardised.

5.3 Status

With the code of 2026-10-06, both on commit 722488a and with the ROM rework of PR #22, the staged NLP does not converge for the LEI-V3 at this wind. Started from the seeds above, the main stage stays close to feasible (primal infeasibility 0.1–0.4) for about 200 iterations, then

diverges and ends in restoration at the 1000-iteration cap. On the last commit the seed itself is harder: the tuner lowered β_{pk} to its floor without bringing the reel-in angle of attack inside its bound. Where the regression since 2026-08-27 comes from (ROM, steering standardisation, seed changes or staging) has not been isolated. Until it has, the seed results of Sect. 5.1 describe the current code, and the optimum of Sect. 5.2 describes what the method produced when it last converged.

6 Reproduction

From the project root:

1. `pythonscripts/reduced-order-model/optimization/cycle/fit_periodic_cycle_config.py--kiteLEI-V3-KITE--auto` (lobe seed), or `--reelin dubins --loops 7 --check` (Dubins seed);
2. `pythonscripts/reduced-order-model/optimization/cycle/run_full_cycle_opti.py --kiteLEI-V3-KITE`;
3. `pythonscripts/reduced-order-model/optimization/cycle/report_cycle_optimisation.py--runlobe=<kitedir>--rundubins=<kitedir>--logs<dir>`, then `latexmk -pdf` in `docs/cycle_optimisation/`.

The seed is written to `<kite>/cycle_configs/full_cycle_periodic_from_exp.yaml` and the optimum to `results/<kitefolder>/optimization/full_cycle/`. To keep the tracked LEI-V3 seed untouched, the runs of Sect. 5 were made on copies of the kite folder; `--kite` accepts any folder laid out like `data/LEI-V3-KITE/`.

References

- [1] O. Cayon, V. van Deursen and R. Schmehl, *Translational dynamics of bridled kites: a reduced-order model in the course reference frame*, Wind Energ. Sci. 11, 1097–1121, 2026.
- [2] O. Cayon and R. Schmehl, trajectory optimisation of soft-kite pumping cycles with a quasi-steady reduced-order model, extended abstract, The Science of Making Torque from Wind (TORQUE), 2026.
- [3] L. E. Dubins, *On curves of minimal length with a constraint on average curvature, and with prescribed initial and terminal positions and tangents*, Am. J. Math. 79, 497–516, 1957.
- [4] F. Monroy-Pérez, *Non-Euclidean Dubins’ problem*, J. Dyn. Control Syst. 4, 249–272, 1998.